TAIM Does Not Transfer to the CICDDoS2019 Benchmark: A Temporal-Detector Mismatch

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Abstract

We test whether a time-aware, adaptive detector transfers to a real public DDoS benchmark. On CICDDoS2019, under strict family and day hold-outs, fold-isolation and validation-calibrated operating points, the adaptive detector TAIM is near-chance on unseen families (ROC-AUC ≈ 0.57, PR-AUC ≈ 0.05, F1 ≈ 0.05), while a supervised Random Forest baseline generalizes strongly (ROC-AUC ≈ 0.99, F1 ≈ 0.79) on 17 held-out families with ≥11 attack windows each. The narrow claim: TAIM does not transfer to this CICDDoS2019 benchmark. We trace the failure to a temporal-detector/benchmark mismatch - CICDDoS2019 family bursts last only minutes, giving ≈3.2 windows per source IP, so there is too little per-device history for a temporal baseline to learn from. We argue this is a benchmark-transfer result, not a deployment-level claim, and that a longer-horizon trace is required for a fair evaluation.

1. Introduction

Network intrusion detectors are usually snapshot methods: they score each observation independently against a model of normal behavior. Time-aware detectors instead build a per-source baseline that updates online, so a sequence of individually-normal windows can accumulate into a sustained-deviation flag (and, in TAIM, escalate a mitigation ladder). The claim is that this buys detection of slow, low-and-slow attacks that a per-window model misses.

That claim is hard to test honestly because the detector is stateful: it must be warmed on training telemetry before it scores anything, and the warm-up must not see the evaluation windows. We therefore run the detector under a strict, fold-isolated protocol on a real public benchmark and ask a narrow question: does adaptive time-aware thresholding remain useful under realistic shift?

2. Background: TAIM and the snapshot baselines

TAIM is a time-aware detector with three components: a per-device baseline that updates online (means/variance per time-of-day slot), a multi-signal fusion gate that flags a window when several signals are simultaneously elevated, and a mitigation ladder that escalates from watch to bandwidth cap to deauth as sustained deviation accumulates. Crucially it is stateful and causal (score-then-update), which is exactly why it must be warmed on training telemetry and scored on frozen state for evaluation.

The baselines are snapshot methods: a supervised Random Forest (log-scaled standardized features), unsupervised Isolation Forest (contamination tuned on inner validation), and a fixed-rule threshold on connection rate or bandwidth. All three score each window independently and need no per-device history.

3. Dataset and strict protocol

Data. CICDDoS2019 (CC-BY-4.0, UNB). The 01-12 and 03-11 captures are streamed, adapted to per-(source-IP, 1-minute) windows, and tagged with their capture family and capture day. We use 1-minute buckets because each family attack burst lasts minutes; coarser buckets collapse a family's attacks into one or two windows. Result: 10,470 windows, 2 capture days, 17 families, 1,275 attack windows; every family has ≥11 attack windows (Syn 669 → WebDDoS 11), and ≈3.2 windows per source IP.

Splits. Never random rows. family hold-out holds out an entire attack family (test = held-out family windows + a 20% benign split held out from training); day hold-out holds out a whole capture day.

Fold isolation + calibration. The adaptive detector is warmed on the training rows only (updates on), then scores the test rows against a frozen baseline (updates off), so held-out family/day telemetry never updates the baseline that scores it. Only the unsupervised contamination is tuned on an inner validation split; the RF decision threshold stays at 0.5. The RF and TAIM recall@FPR cutoffs are each selected on a genuine validation split (RF: a fresh fit-split/validation-split model; TAIM: a chronological train-timeline split where the baseline is warmed on the earlier fraction and the cutoff is picked on the later, frozen-scored fraction). The test fold is never touched for calibration. Comparators see the same 5 windowed signals (bandwidth_mbps, conn_rate_ps, port_div, pkt_size_mean, app_req_ps; bandwidth/app_req log-scaled).

3.1 Methodological hardening

Three evaluation habits were corrected before this result was trusted. First, fold isolation: a stateful detector cannot be run over the concatenated train+test frame and then have its test rows read, because its adaptive baseline would have processed the evaluation telemetry. TAIM is therefore run two-phase, warming on train and scoring test frozen. Second, in-sample calibration: a recall@FPR cutoff must be selected on a validation portion, not on the sequence whose scores are reported. Third, per-family support: a family-level result is only meaningful when each held-out family has enough attack windows, which required 1-minute buckets (coarser buckets left most families with a handful of windows). These are the standard pitfalls for any online detector.

4. Results

Strict family hold-out (17 held-out families, ≥11 attack windows each):

comparator	F1 (±95% CI)	prec.	recall	PR-AUC (±CI)	ROC-AUC (±CI)	recall@1% FPR	alerts
Random Forest (sup.)	**0.792 (±0.049)**	**0.742**	**0.902**	**0.956 (±0.034)**	**0.992 (±0.010)**	**0.919**	**≈5**
Isolation Forest	0.117 (±0.028)	0.105	0.856	—	—	—	≈9
Fixed rule	0.090 (±0.063)	0.053	0.976	—	—	—	≈11
TAIM (adaptive)	0.052 (±0.032)	0.029	0.541	0.046 (±0.031)	0.569 (±0.107)	0.540	≈81

Per held-out family, supervised Random Forest F1 ranges 0.52–0.90 and ROC-AUC 0.92–1.00 across all 17 families (see Table 2); TAIM F1 is 0.00–0.23. Under strict day hold-out (2 capture days), RF F1 ≈ 0.70 (AUC ≈ 0.92) versus TAIM F1 ≈ 0.18.

Important operating-point caveat. The 1% FPR cutoff is a *validation-selected* threshold, not a guaranteed operational false-positive rate. When applied to the held-out test folds, the TAIM cutoff produced an achieved test FPR of ≈ 0.395 - over 39 times the 1% budget - because TAIM's score does not separate attacks under shift. The corresponding RF cutoff achieved FPR ≈ 0.005. These are test-set FPRs observed under shift, not a guarantee that the detector will operate at 1% FPR in deployment.

Table 2. Per held-out family, supervised Random Forest (fold-isolated):

held-out family	attack windows	F1	ROC-AUC
DrDoS_DNS	36	0.875	0.996
DrDoS_UDP	46	0.900	0.998
UDP-lag	41	0.901	1.000
DrDoS_SNMP	30	0.853	1.000
MSSQL	28	0.844	0.998
Portmap	79	0.838	0.984
DrDoS_LDAP	12	0.688	0.995
DrDoS_NTP	172	0.521	0.921
Syn	669	0.749	0.975
WebDDoS	11	0.710	1.000

Minimum family support is 11 attack windows (WebDDoS); the well-sampled families reach F1 0.9, and even the hardest (DrDoS_NTP) reaches AUC 0.92. The range is stable, so the supervised result is not an artifact of one easy family.

4.1 Why TAIM is near-chance here

TAIM is temporal: it learns a per-device baseline over time and flags sustained deviation. On CICDDoS2019 the family bursts are short, giving roughly 3.2 windows per source IP, so there is essentially no per-device history for TAIM's baseline to learn from and no time-of-day regime to model. Even after fold isolation and validation calibration, TAIM's score orders attacks about as well as chance: its validation-calibrated recall@1%FPR cutoff achieved a test FPR of ≈ 0.395 on held-out folds, ≈39 times the 1% budget, i.e. the score does not separate the classes. Isolation Forest and Random Forest are snapshot models that need no such history, so they are not similarly handicapped. We read this as a benchmark mismatch, not a general failure of temporal detection.

5. Related work

Online, time-aware detection is a common framing for network intrusion detection, and adaptive baselining is attractive because it does not require labeled attacks. Empirical work on public DDoS/IDS benchmarks (CIC-IDS, CSE-CIC-IDS, UNSW-NB15) most often reports snapshot methods. Less reported is whether the *temporal* premise survives when the benchmark's attacks are short and family-isolated rather than spread over a long host history. Our result is a concrete counterexample: a temporal detector that is strong on long-horizon synthetic traffic is near-chance on a real benchmark whose per-device history is too short. This is the same synthetic-to-real gap documented for host anomaly detectors, observed here for an online network detector.

6. Conclusion

On real CICDDoS2019 with strict family/day hold-outs, fold isolation and validation calibration, adaptive time-aware thresholding (TAIM) does not remain useful under distribution shift: it is near-chance on unseen families while a supervised baseline generalizes strongly. The result is a benchmark-transfer finding: TAIM's per-device temporal baseline has no purchase on short, family-isolated bursts. A longer-horizon real trace (e.g. LANL or a multi-day capture) is the correct next test before any deployment claim.

7. Limitations

This is a bounded per-family sample, not the full 22 GB trace. Only two capture days are available, so the day-hold-out confidence interval is wide and the family hold-out (n = 17) is the primary evidence.

TAIM's internal state is not re-run deterministic across separate process invocations (pre-existing), so the reported aggregates use a fixed evaluation pass. The 1-minute bucket also inflates absolute per-family window counts but does not change the ranking. These caveats do not change the result: the adaptive detector falls far behind a supervised baseline even under a fold-isolated, calibrated protocol.

8. Repository, data, and reproduction

Repository (anonymized for review): github.com/Farooq-Syed/taim. The evaluation entry point is src/real_cicddos_eval.py; the window builder is scripts/build_real_windows.py; the capture downloader is scripts/download_real_data.py. Frozen preprocessing and split definitions are documented in REAL_DATA_RESULTS.md.

Data attribution and license. CICDDoS2019 (CC-BY-4.0), Canadian Institute for Cybersecurity, University of New Brunswick. Dataset page: https://www.unb.ca/cic/datasets/ddos-2019.html. The captures are public and used here for research evaluation only.

Reproduction commands.

python scripts/download_real_data.py; python scripts/build_real_windows.py --rows-per-family 1000000 --bucket-min 1 --output data/cicddos_real_windows.csv; python src/real_cicddos_eval.py --input data/cicddos_real_windows.csv --split family; python src/real_cicddos_eval.py --input data/cicddos_real_windows.csv --split day

9. AI-use disclosure

AI coding assistance was used during implementation and drafting. The author directed the research question, the benchmark evaluation protocol, the strict split and calibration design, the interpretation of the negative result, and reviewed and verified the final code and manuscript claims. AI assistance did not set the research direction or the claims.

References

[1] CICDDoS2019 dataset, Canadian Institute for Cybersecurity, University of New Brunswick. https://www.unb.ca/cic/datasets/ddos-2019.html

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[3] Liu, Ting & Zhou, Isolation Forest, ICDM 2008. Breunig et al., LOF, SIGMOD 2000. Breiman, Random Forests, ML 2001.

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Appendix. Three review questions

1. Is the claim narrow enough? Yes: 'adaptive time-aware thresholding does not transfer to this benchmark' - explicitly a benchmark-transfer result, not a deployment-level claim.

2. Is the split protocol sound? Family and day hold-outs are strict; the adaptive detector is fold-isolated (warmed on train, scored frozen); the recall@FPR cutoffs are calibrated on validation, never the test fold.

3. What experiment would most change your confidence? A longer-horizon real trace (e.g. LANL or a multi-day capture) that gives each source IP enough history for a temporal baseline to learn from - the exact condition this benchmark lacks.